Peripherals and Interfacing

Course Code: CSE-3207

Topics: Keyboard & Mouse



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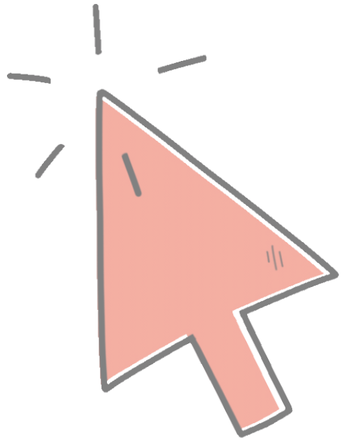
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What is Computer Peripherals?

A **peripheral device** is an auxiliary device used to put information into and get information out of a computer. They are optional unlike hardware.

List of Some Common Peripherals

Input	Output	Storage	Others
Keyboard, Mouse , Camera, Microphone, Touch Screen, Scanner, Barcode Reader	CRT Monitor, Plasma Display, Video Projector, Printer, Plotter, Speaker	Magnetic Disk, Compact Disk, DVD, Flash Memory	Modem, Display Adaprer, Sound Card

What is a Computer Keyboard?

A **keyboard** is one of the primary input devices used with a computer partially modeled after the typewriter keyboard. Keyboards are designed for the input of text and also to control the operation of a computer.

How a Keyboard Basically Works?

- When a key is pressed, it pushes down on a rubber dome sitting beneath the key.
- A conductive contact on the underside of the dome touches (and hence connects) a pair of conductive lines on the circuit below.
- This allows current to flow, changing the signal strength.
- A scanning signal is emitted by the chip along the pairs of lines to all the keys. When the signal in one pair becomes different, the chip generates a "make code" corresponding to the key connected to that pair of lines.
- The code generated is sent to the computer either via a keyboard cable or over a wireless connection.
- A chip inside the computer receives the signal bits and decodes them into the appropriate keypress.
- When the key is released, a break code (different than the make code) is sent to indicate the key is no longer pressed.

Basic Internal Architecture of a Keyboard

A keyboard is a lot like a miniature computer. It has its own processor and circuitry that carries information to and from that processor. A large part of this circuitry makes up the **key matrix**.



Fig 01: The microprocessor and controller circuitry of a keyboard

What is Key Matrix?

- A key matrix is a grid of circuits underneath the keys.
- When a key is pressed, it presses a switch, completing the circuit and allowing a tiny amount of current to flow through.
- When the processor finds a circuit that is closed, it compares the location of that circuit on the key matrix to the **character map** in its read-only memory (ROM).

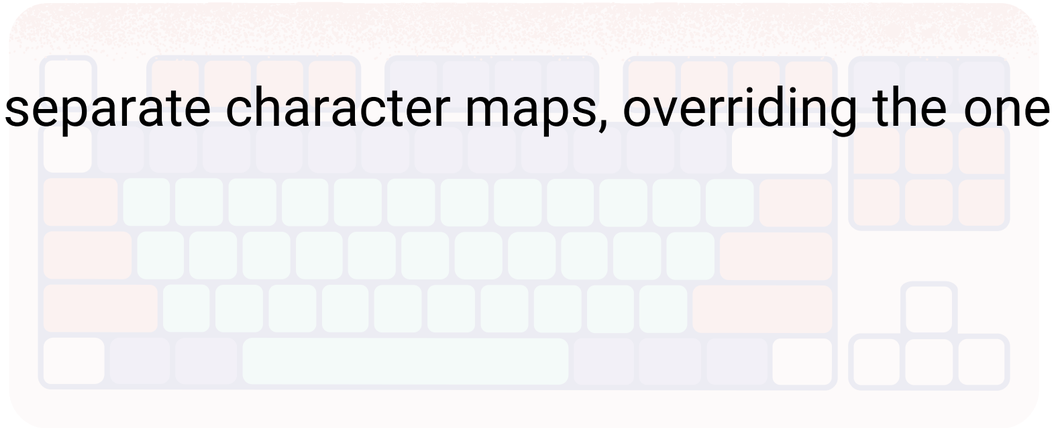


Fig 02: The key matrix

What is Character Map?

A **character map** is basically a comparison chart or lookup table. It tells the processor the position of each key in the matrix and what each keystroke or combination of keystrokes represents.

A computer can also use separate character maps, overriding the one found in the keyboard.



Keyboard Types

There are many types of keyboard, usually differentiated by the switch technology employed in their operation. The choice of switch technology affects key response and travel.



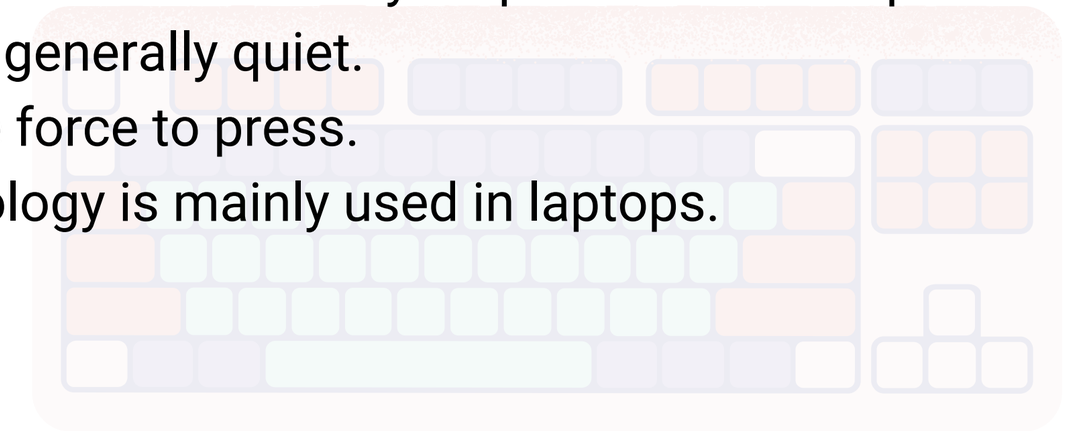
- Dome-switch keyboard
- Scissor-switch keyboard
- Capacitive keyboard
- Mechanical-switch keyboard
- Hall-effect keyboard
- Laser keyboard
- Membrane keyboard

Dome-switch Keyboard

- They use small, flexible rubber domes, each with a hard carbon center.
- When you press a key, a plunger on the bottom of the key pushes down against the dome.
- The carbon center presses against a hard, flat surface beneath the key matrix.
- As long as the key is held, the carbon center completes the circuit.
- When the key is released, the rubber dome springs back to its original shape, forcing the key back up to its at-rest position.
- These are inexpensive, have pretty good tactile response.
- Used in handheld controllers, mobile phones, automotive, consumer electronics and medical devices.

Scissor-switch Keyboard

- Scissor-switch technology does not use a rubber-plunger assembly like most other keyboards.
- Scissor-switch keyboards are normally responsive and crisp.
- These keyboards are generally quiet.
- The keys require little force to press.
- This keyboard technology is mainly used in laptops.

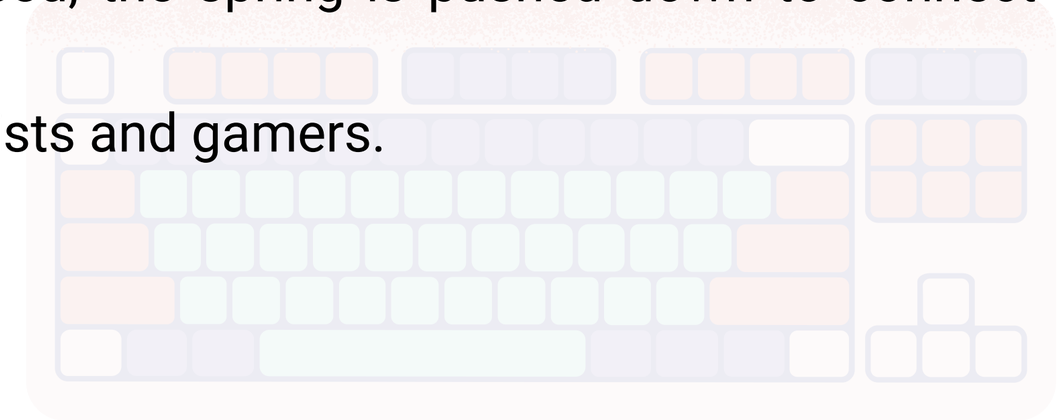


Capacitive Keyboard

- Unlike traditional keyboards, where electricity flow initiates internally when a key is pressed, electricity is always flowing in capacitive keyboard.
- Under each key, static charges are stored in capacitors. When a key is pressed, it connects with the capacitor pads and changes the capacitance at that specific point on the capacitor pad, which the keyboard identifies and registers as the exact keystroke/key.
- Capacitive keyboards are faster than contact keyboards as they only require a gentle push to register a keystroke.
- Capacitive keyboards are expensive, and resist wear, water, foreign objects and dirt.
- They are common in PC keyboards.

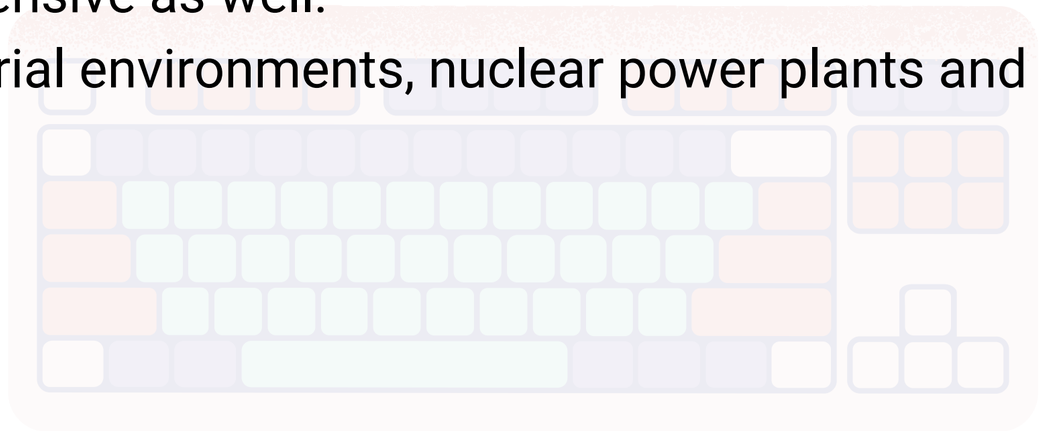
Mechanical Keyboard

- These keyboards don't use the rubber dome system.
- Instead, they have switches under the keys which have springs in them.
- When a key is pressed, the spring is pushed down to connect the circuit and register a keystroke.
- Users are mainly typists and gamers.

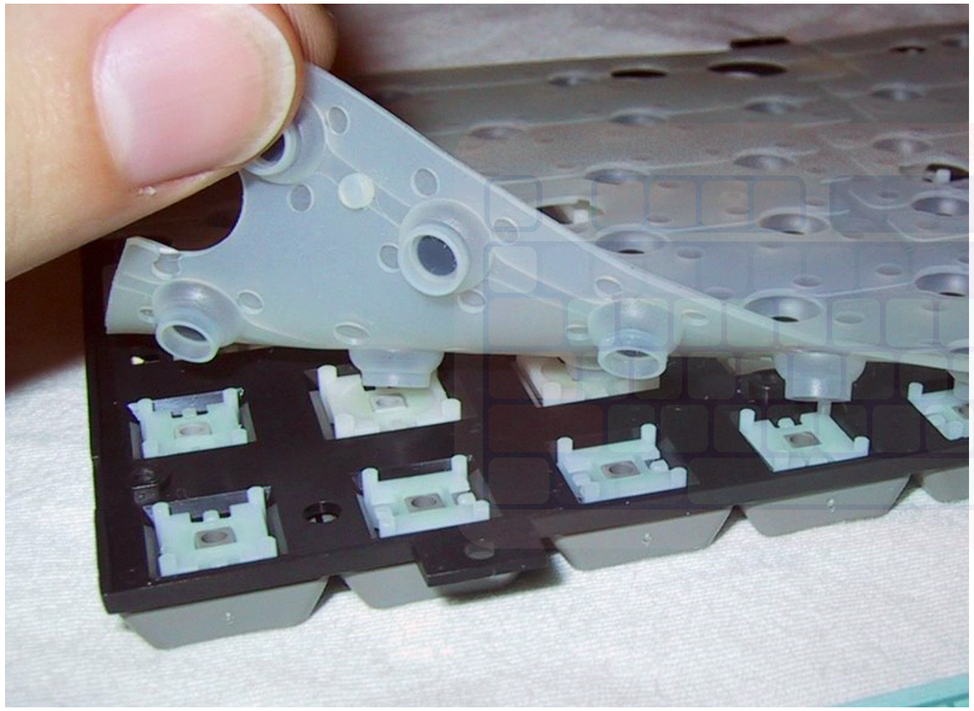


Hall-effect Keyboard

- Hall effect keyboards use magnets and Hall effect sensors instead of an actual switch.
- When a key is depressed, it moves a magnet, which is detected by the solid-state sensor.
- Very reliable and expensive as well.
- Mainly used in industrial environments, nuclear power plants and aircraft cockpits etc.

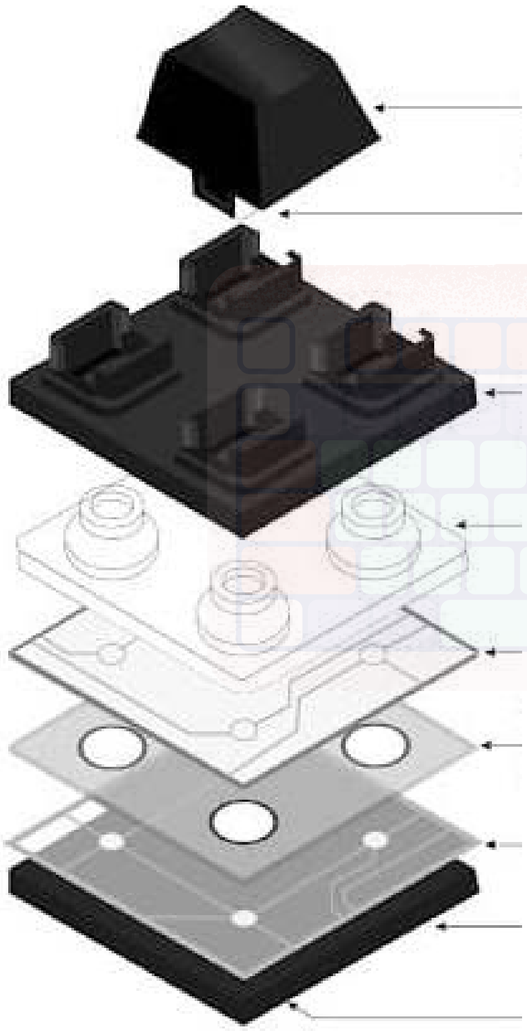


Membrane Keyboard



- Quieter
- Cheaper
- Softer typing feel

Layers of a Computer Membrane Keyboard



Keycap

Integrated Slider

Upper case

Rubber domes

Top membrane layer

Centre membrane layer

Bottom membrane layer

Lower Case

Ca 0.06-0.09 mm membrane thickness

Laser Keyboard





A keyless keyboard and uses holograms instead of solid keys for keystrokes.

Also called Projection Keyboard.

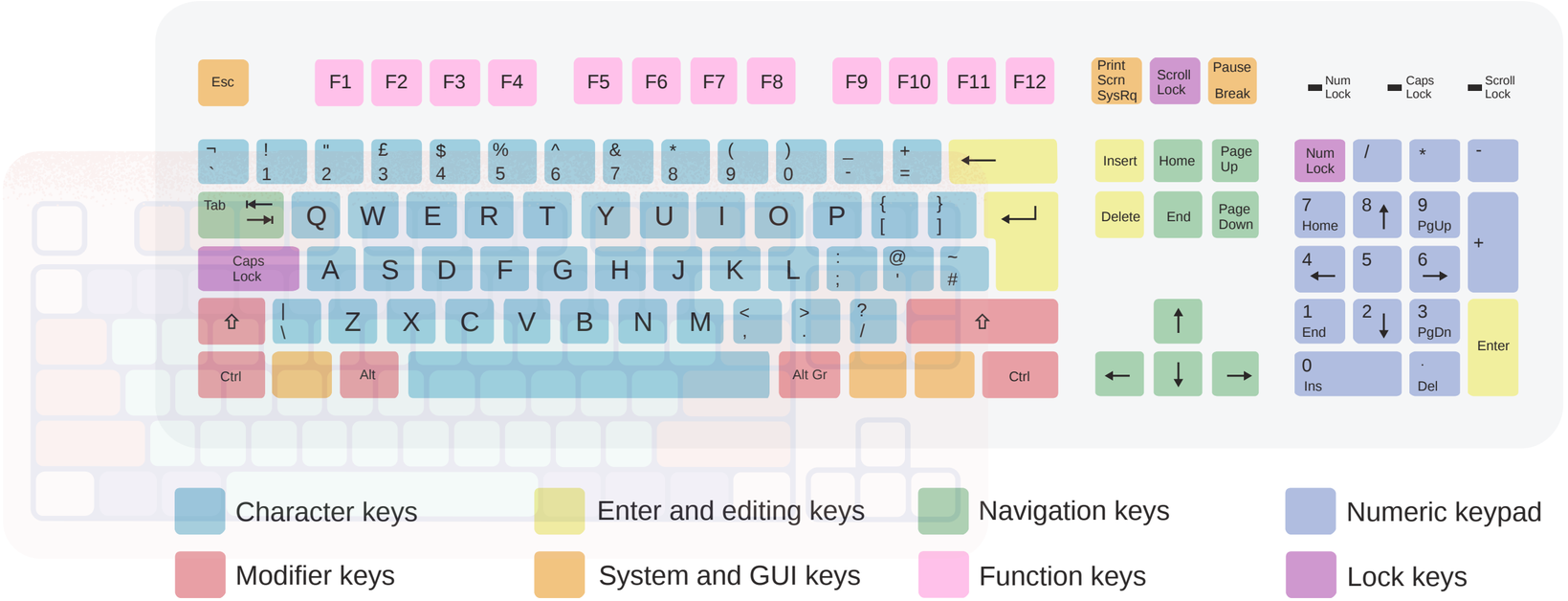
Working principles of Laser Keyboard



- **A CMOS (complimentary metal-oxide semiconductor) images the finger's position within the area of the keyboard.**
- **A special sensor chip called a Virtual Interface Processing Core analyzes the location of the intended keystroke.**
- **Finally the device sends the signal to PC.**

Keyboard Layout

- **QWERTY**
- **DVORAK**
- **COLEMAK**



A typical 105-key computer keyboard

Different standard QWERTY keyboard

101/104 - ANSI



101/104 *variant*



102/105 - ISO



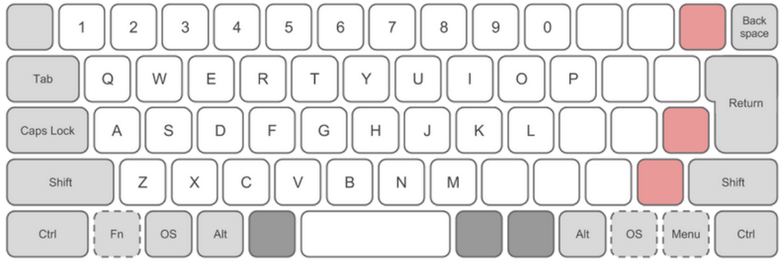
103/106 - KS



104/107 - ABNT



106/109 - JIS



Keyboard connection types

- **Wired Keyboard**
    1. PS/2 type keyboard connector
    2. USB type keyboard connector

- **Wired Keyboard**
    1. Bluetooth keyboard connector
    2. Wi-Fi standard keyboard connector
    3. Infrared(IR) keyboard connector
    4. Radio Frequency(RF) keyboard connector

A PS/2 type keyboard connector

1



A USB type keyboard connector

2





Laser Keyboard

3



Bluetooth connected Keyboard

4



Customization of keyboard

Under systems running X11 (e.g. GNU/Linux) this can be done with

- xmodmap

Windows has several software for this purpose :

- SharpKeys: free
- KeyTweak: free
- Keyboard Layout Manager: commercial

Input Device: MOUSE



Computer Mouse

In computing, a mouse (plural mice or mouses) functions as a pointing device by detecting two- dimensional motion relative to its supporting surface.

How does a computer mouse work?

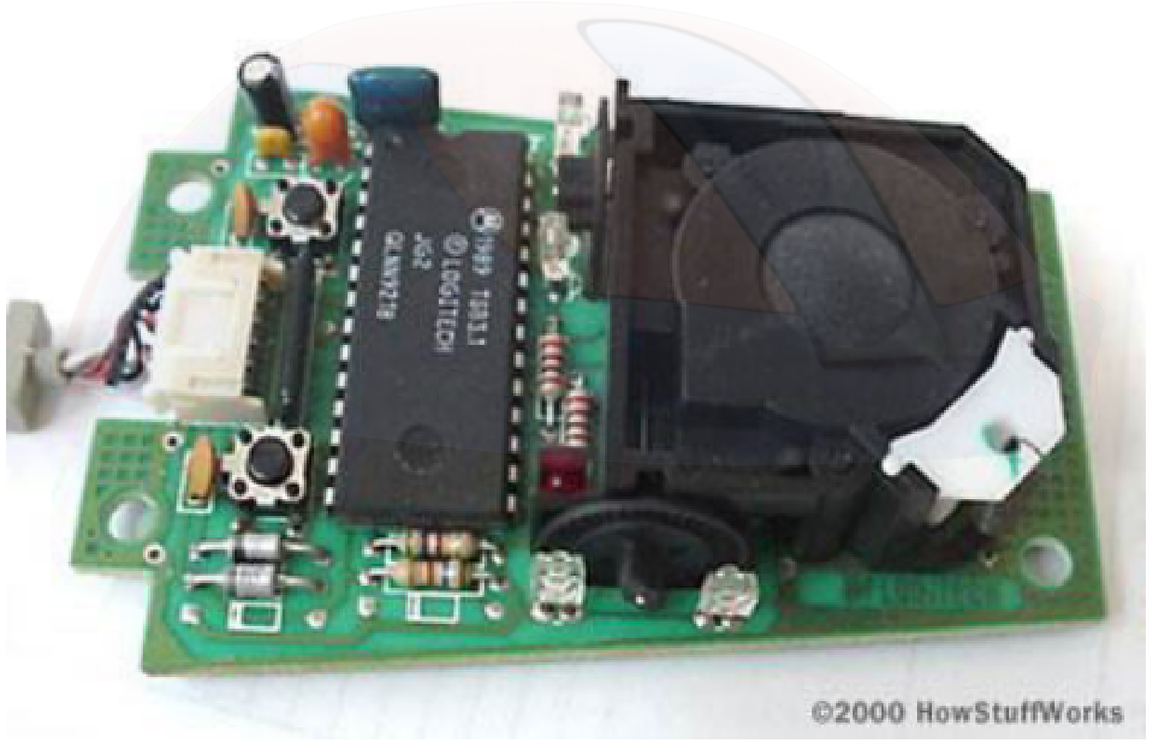
The first types of mice were mostly mechanical, a ball rolled around under the body of the mouse, and rollers sensed the direction and speed at which the mouse was moved.

The rollers were connected to a disc that contained holes and used an IR LED (Light Emitting Diode) to sense the movement.

Two sensors per wheel gave both speed and directional information to an on-board integrated circuit that decoded this data and transmitted it to the computer.

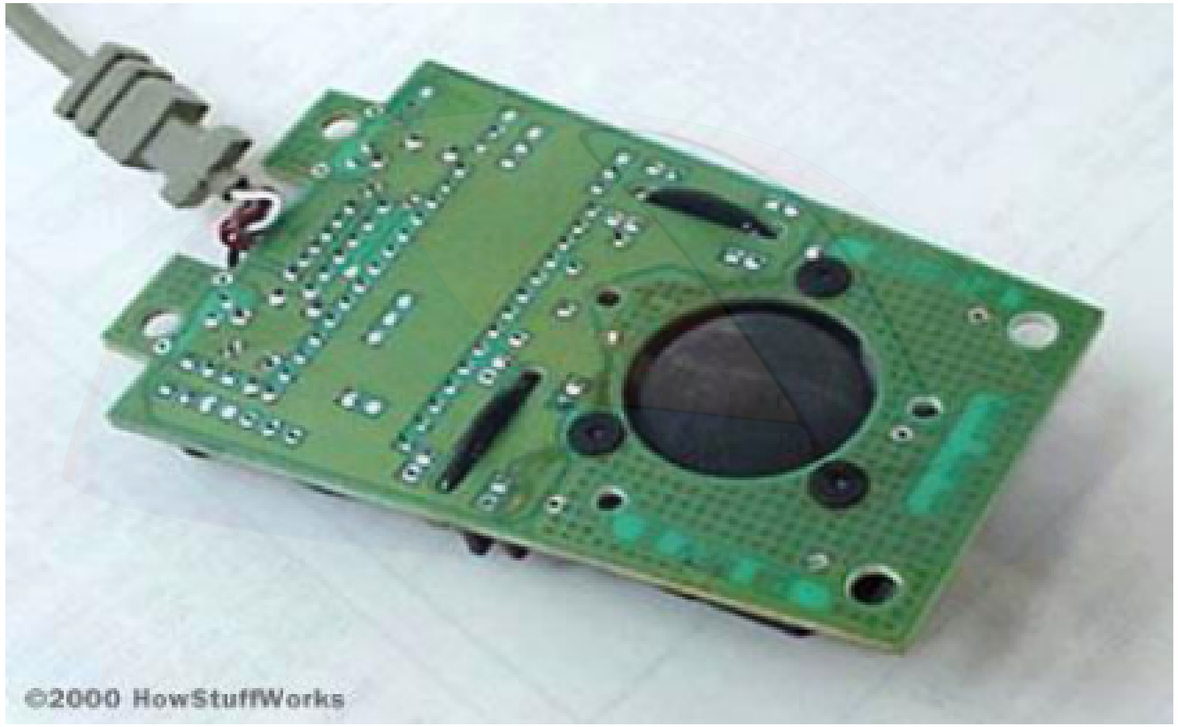
Main Goal

The main goal of any mouse is to translate the motion of your hand into signals that the computer can use.



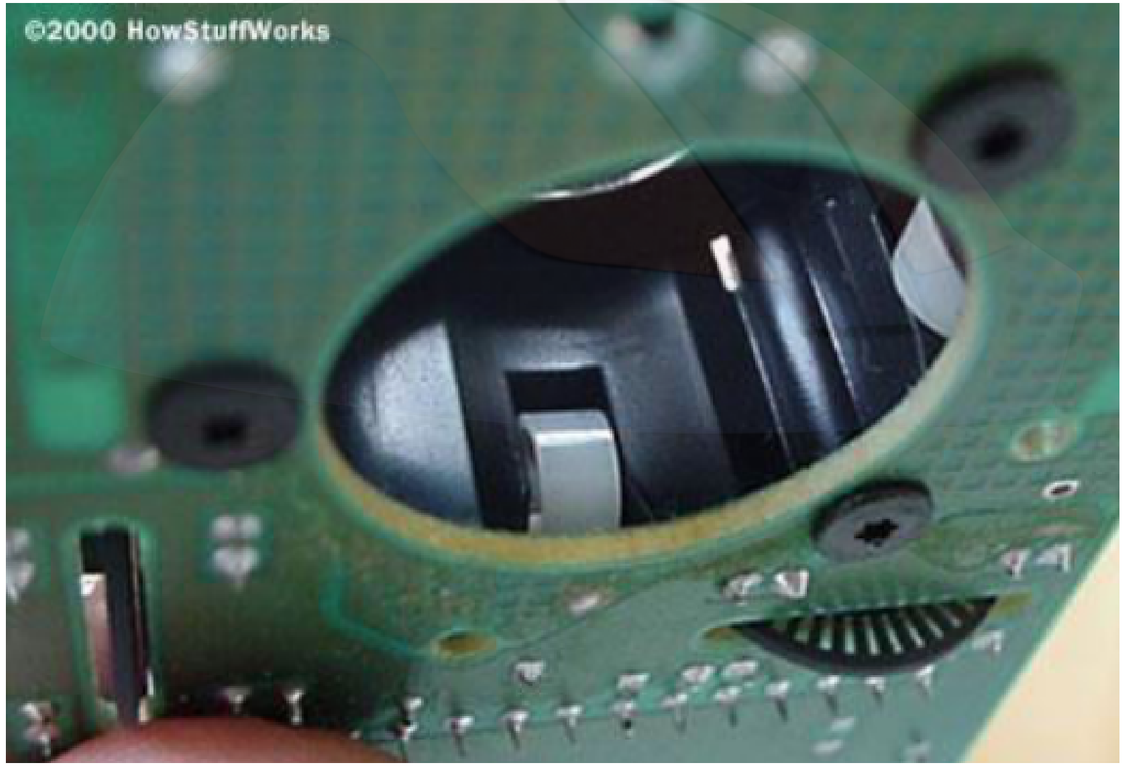
Working Process of Mouse

A ball inside the mouse touches the desktop and rolls when the mouse moves.



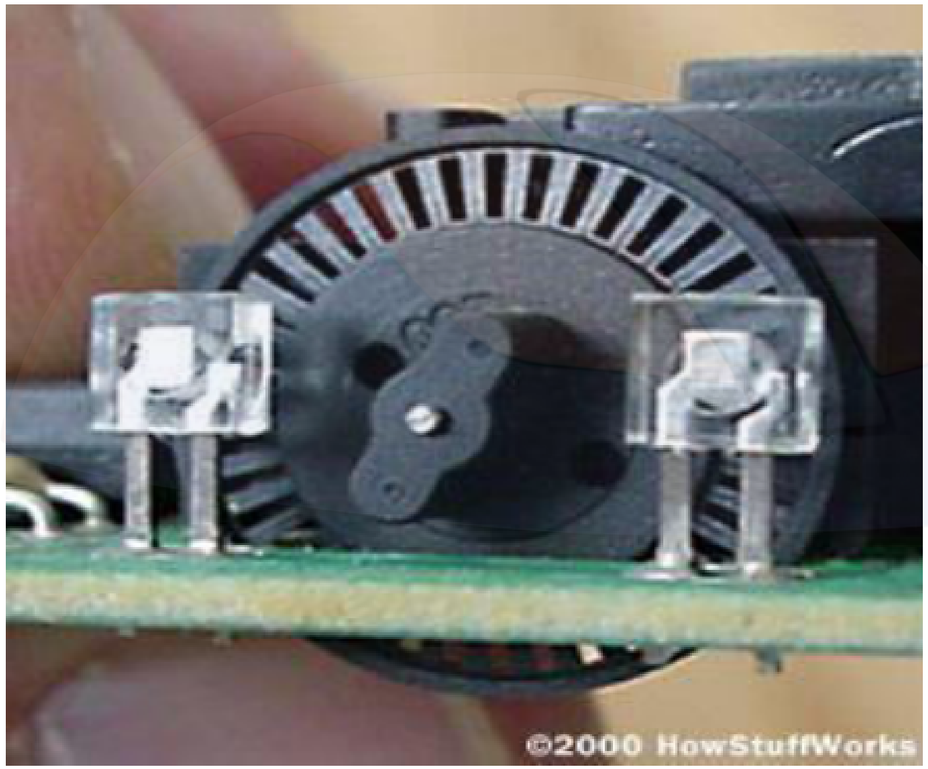
Working Process of Mouse

Two rollers inside the mouse touch the ball. One of the rollers is oriented so that it detects motion in the X direction, and the other is oriented 90 degrees to the first roller so it detects motion in the Y direction.



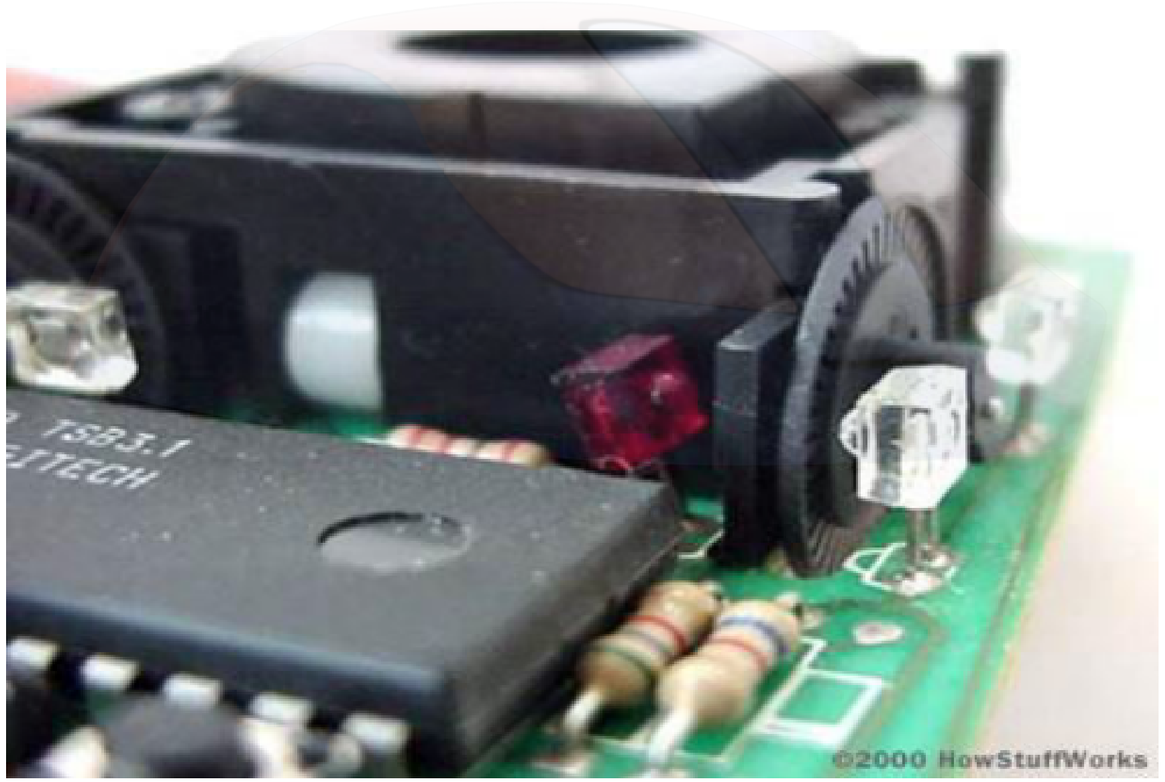
Working Process of Mouse

The rollers each connect to a shaft, and the shaft spins a disk with holes in it. When a roller rolls, its shaft and disk spin.



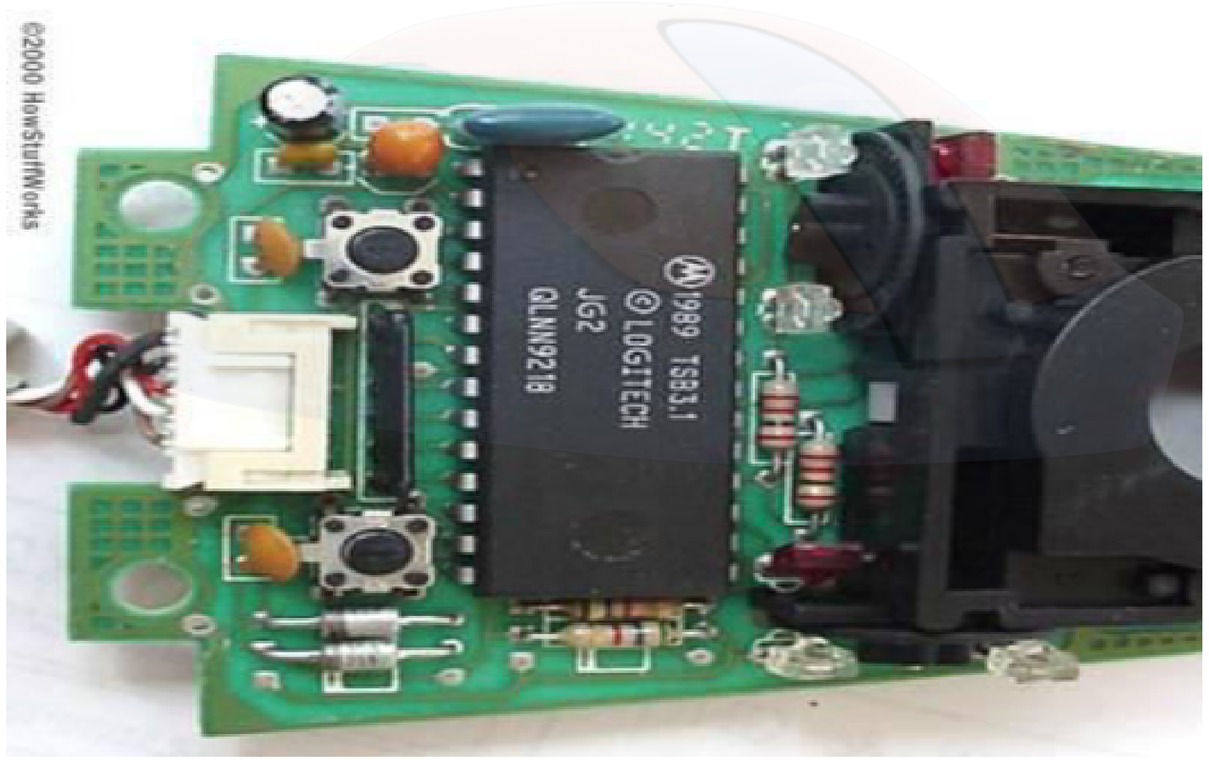
Working Process of Mouse

On either side of the disk there is an infrared LED and an infrared sensor. The holes in the disk break the beam of light coming from the LED so that the infrared sensor sees pulses of light. The rate of the pulsing is directly related to the speed of the mouse and the distance it travels.



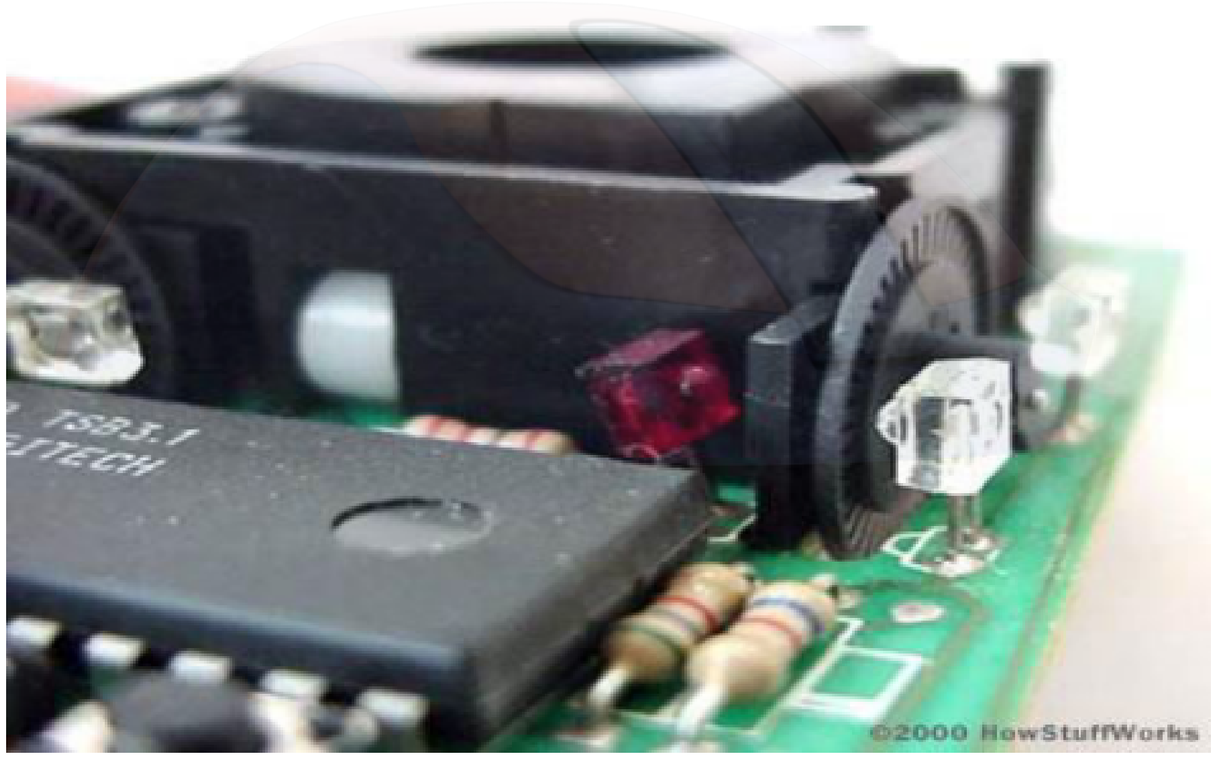
Working Process of Mouse

An on-board processor chip reads the pulses from the infrared sensors and turns them into binary data that the computer can understand. The chip sends the binary data to the computer through the mouse's cord.



Working Process of Mouse

Each encoder disk has two infrared LEDs and two infrared sensors, one on each side of the disk (so there are four LED/sensor pairs inside a mouse). This arrangement allows the processor to detect the disk's direction of rotation.



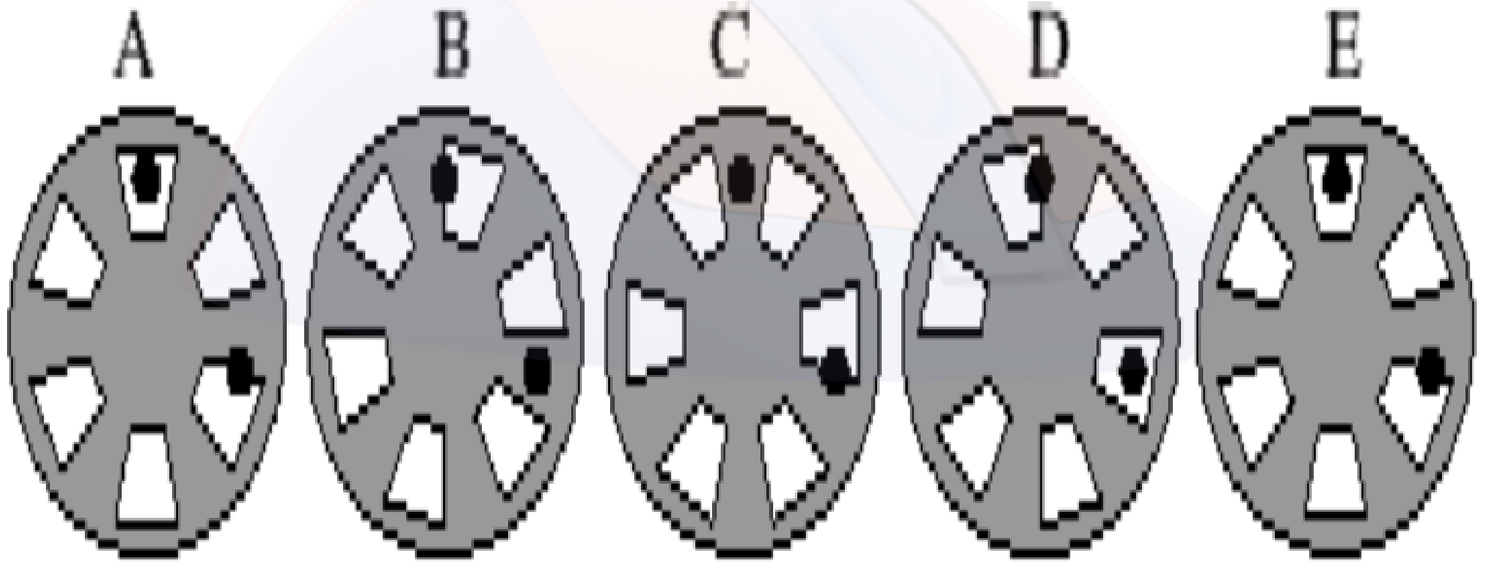
Working Process of Mouse

Open up a mouse and inside it you will find two wheels, each one similar to the first drawing. The wheel is usually made of black plastic with rectangular slots punched in it.

Each LED shines on to a light sensitive transistor. The two emitters are spaced so that, when one transistor can 'see' its LED through the centre of its window, the other LED is looking at an edge and is therefore switching on or off. In my illustration the LEDs are spaced at 105° (60° x 1.75). The output voltage from the transistor is processed to switch rapidly from high to low as the LED's light is transmitted or occluded so that the voltage is low when the transistor is lit and high when it is in darkness. In the diagram LED A is fully illuminated and LED B is switching. Note that LED B may be switching from light to dark or from dark to light - this depends on the rotation direction.

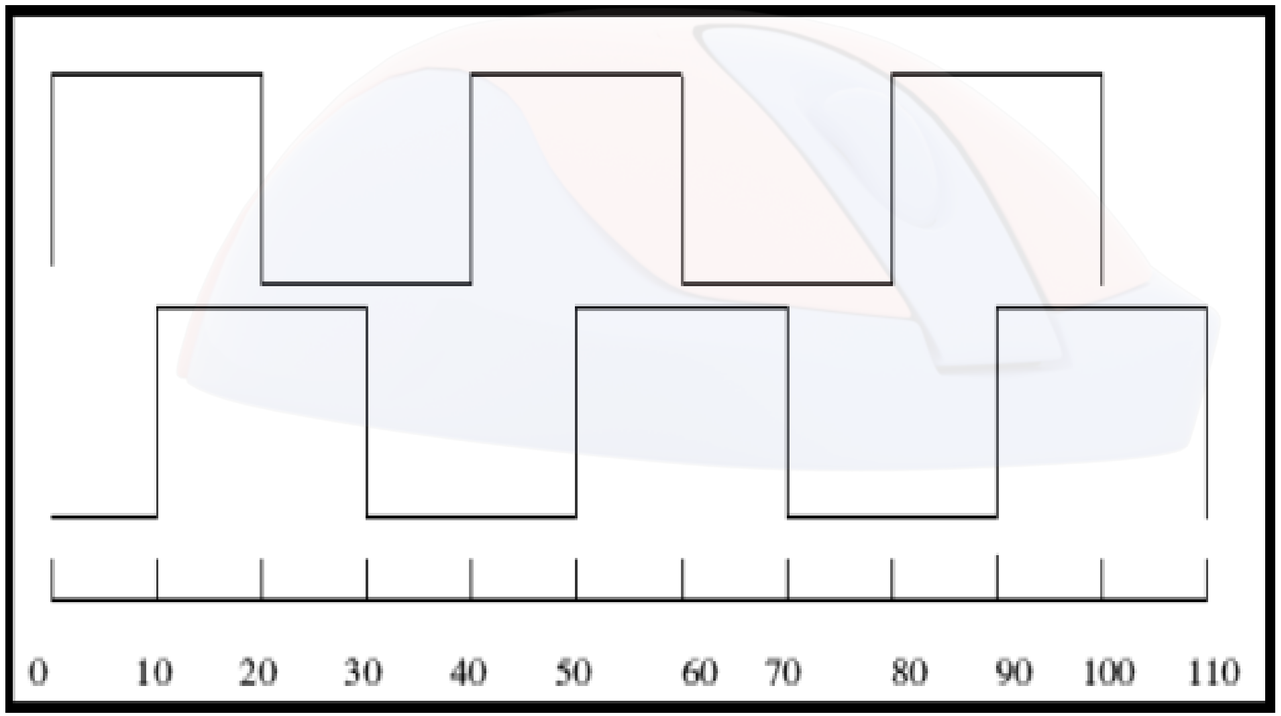
Working Process of Mouse

Here the wheel is shown in 4 different states, each 15° rotated from the last. Diagram E is equivalent to diagram A, being 60° rotated. For clockwise rotation the states follow each other in order A-B-C-D-E from left to right but if you read the states from right to left, E-D-C-B-A, then these correspond to anticlockwise rotation



Working Process of Mouse

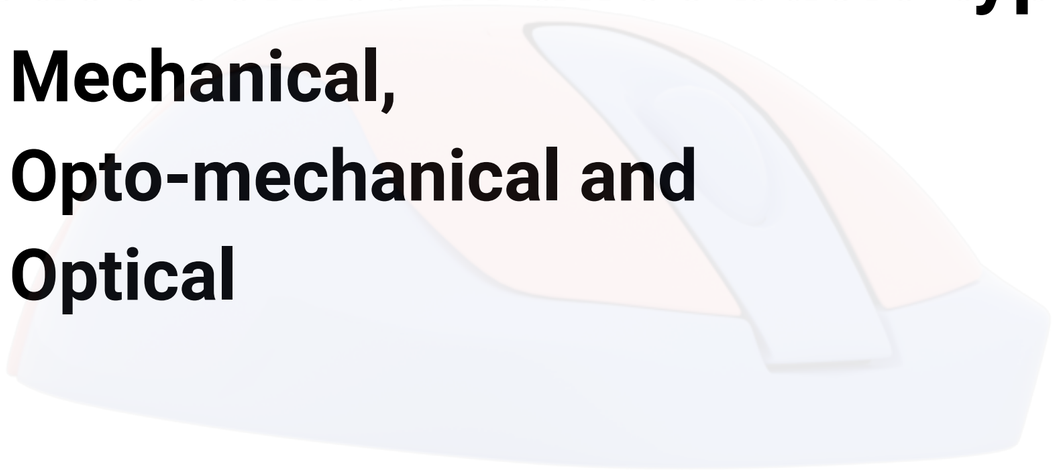
The diagram below shows the corresponding electrical signals switching at 15° intervals There are two such wheels, one rotates for vertical movement and the other rotates for horizontal movement of the mouse ball



Varieties of Mice

The mouse comes in three basic types:

1. **Mechanical,**
2. **Opto-mechanical and**
3. **Optical**



Mechanical Mouse

Operating of a Mechanical Mouse:

1: Moving the mouse turns the ball.

2: X and Y rollers grip the ball and transfer movement.

3: Optical encoding disks include light holes.

4: Infrared LEDs shine through the disks.

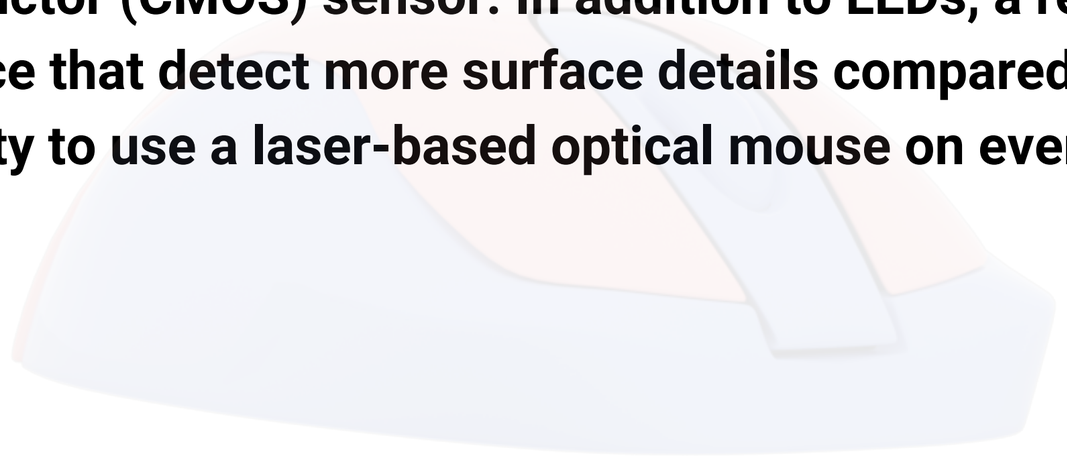
5: Sensors gather light pulses to convert to X and Y velocities.



Optical Mouse

The optical mouse actually uses a tiny camera to take thousands of pictures every second.

Able to work on almost any surface without a mouse pad, most optical mice use a small, red light-emitting diode (LED) that bounces light off that surface onto a complimentary metal-oxide semiconductor (CMOS) sensor. In addition to LEDs, a recent innovation is laser-based optical mice that detect more surface details compared to LED technology. This results in the ability to use a laser-based optical mouse on even more surfaces than an LED mouse.



Optical Mouse

The CMOS sensor sends each image to a digital signal processor (DSP) for analysis.

The DSP detects patterns in the images and examines how the patterns have moved since the previous image.

Based on the change in patterns over a sequence of images, the DSP determines how far the mouse has moved and sends the corresponding coordinates to the computer.

The computer cursor moves the cursor on the screen based on the coordinates received from the mouse. This happens hundreds of times each second, making the cursor appear to move very smoothly.

Accuracy of Optical Mouse

A number of factors affect the accuracy of an optical mouse. One of the most important aspects is resolution. The resolution is the number of pixels per inch that the optical sensor and focusing lens "see" when you move the mouse. Resolution is expressed as dots per inch (dpi). The higher the resolution, the more sensitive the mouse is and the less you need to move it to obtain a response.



Optical vs Mechanical Mouse

Optical mice have several benefits over track-ball mice:

- No moving parts means less wear and a lower chance of failure.
- There's no way for dirt to get inside the mouse and interfere with the tracking sensors.
- Increased tracking resolution means a smoother response.
- They don't require a special surface, such as a mouse pad.

Optical vs Mechanical Mouse

Optical Mouse generally cannot track on glossy and transparent surfaces, including some mouse pads, sometimes causing the cursor to drift unpredictably during operation.

Mechanical mice have lower average power demands than their optical counterparts. This typically has no practical impact on users of cabled mice (except possibly those used with battery-powered computers, such as notebook models), but has an impact on battery-powered wireless models.

Wireless Mouse

A Cordless or Wireless mouse no longer has to be physically connected to the computer. Two different technologies are used. Some mice operate by an infrared port, where the mouse must have a clear line of sight to the computer, much like a television remote control. Other mice use radio frequencies for their wireless technology, which allows the mice to roam farther from the computer. Most wireless mice use radio frequency (RF) technology to communicate information to your computer.

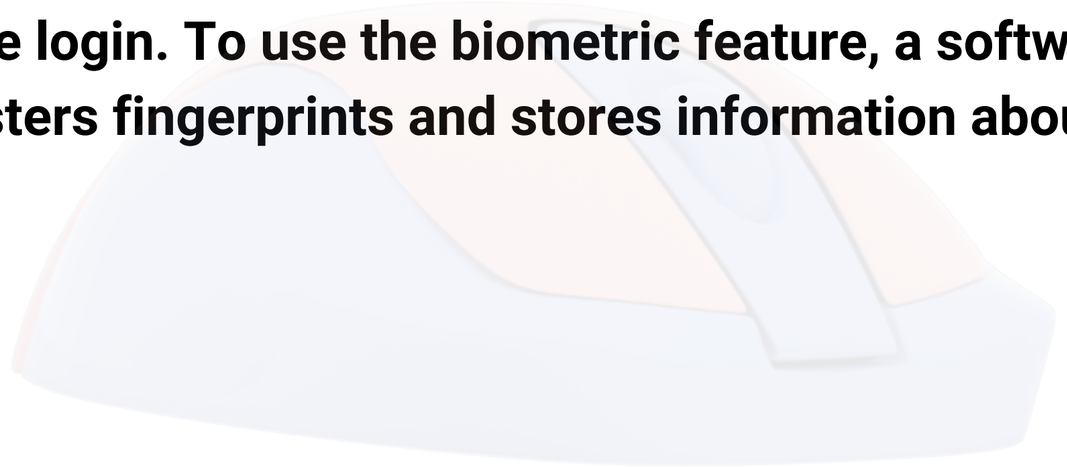


Bluetooth Mouse

One of the RF technologies that wireless mice commonly use is Bluetooth. Bluetooth technology wirelessly connects peripherals such as printers, headsets, keyboards, and mice to Bluetooth-enabled devices such as computers and personal digital assistants (PDAs). Because a Bluetooth receiver can accommodate multiple Bluetooth peripherals at one time, Bluetooth is also known as a personal area network (PAN). Bluetooth devices have a range of about 33 feet (10 meters). Bluetooth operates in the 2.4 GHz range using RF technology. It avoids interference among multiple Bluetooth peripherals through a technique called spread-spectrum frequency hopping

Biometric Mouse

Biometric mice add security to your computer system by permitting only authorized users to control the mouse and access the computer. Protection is accomplished with an integrated fingerprint reader either in the receiver or the mouse. This feature enhances security and adds convenience because you can use your fingerprint rather than passwords for secure login. To use the biometric feature, a software program that comes with the mouse registers fingerprints and stores information about corresponding authorized users



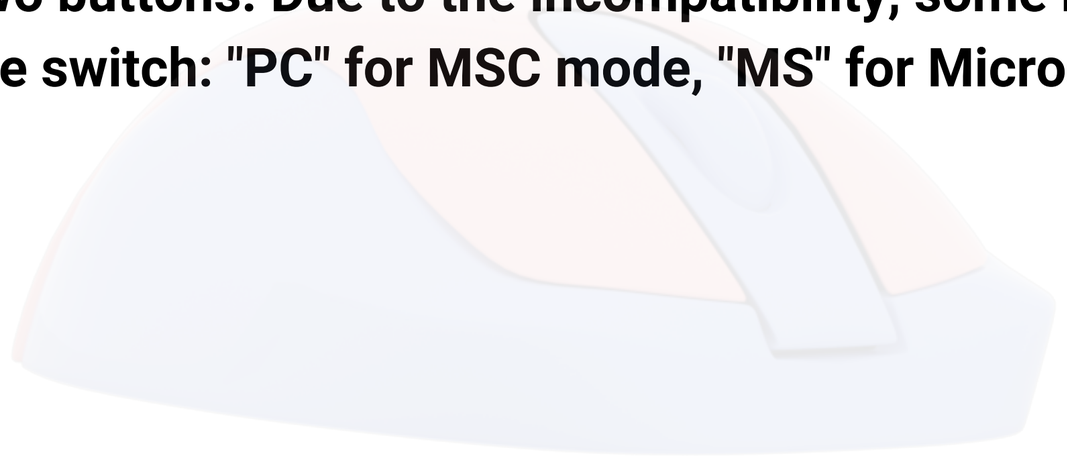
Connectivity and communication protocol

To transmit their input, typical cabled mice use a thin electrical cord terminating in a standard connector, such as

- **RS-232C,**
- **PS/2,**
- **ADB**
- **USB.**

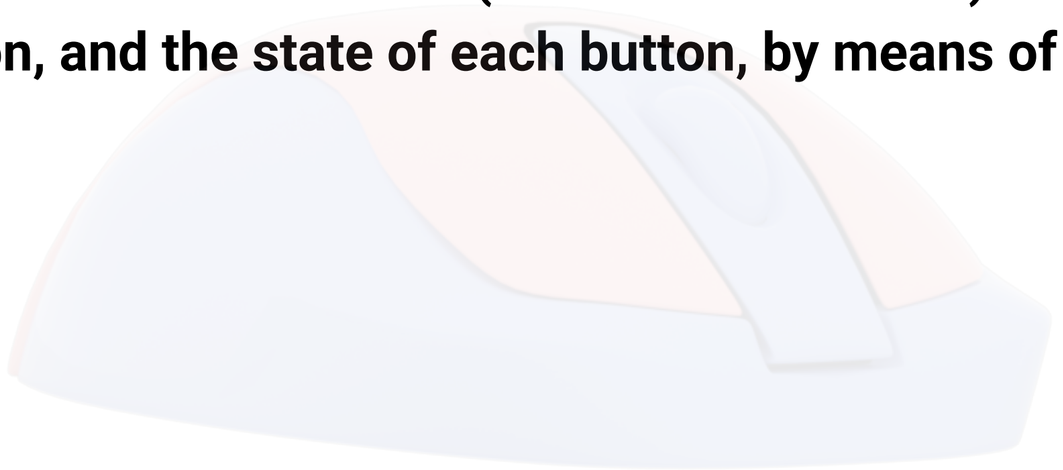
Serial Interface and protocol

Standard PC mice once used the RS-232C serial standard (released in 1969), via a DB-9 connector. The Mouse Systems Corporation version used a five-byte protocol and supported three buttons. The Microsoft version used an incompatible three-byte protocol and only allowed for two buttons. Due to the incompatibility, some manufacturers sold serial mice with a mode switch: "PC" for MSC mode, "MS" for Microsoft mode.



PS/2 Interface and protocol

With the arrival of the IBM PS/2 personal-computer series in 1987, IBM introduced the eponymous PS/2 interface for mice and keyboards, which other manufacturers rapidly adopted. The most visible change was the use of a round 6-pin mini-DIN, in lieu of the former 5-pin connector. In default mode (called stream mode) a PS/2 mouse communicates motion, and the state of each button, by means of 3-byte packets.



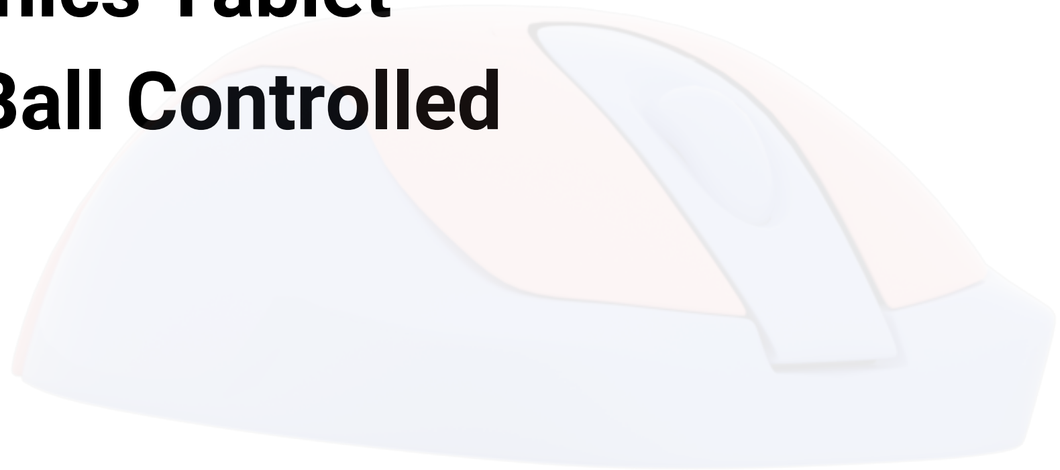
PS/2 Interface and protocol

Here, XS and YS represent the sign bits of the movement vectors, XV and YV indicate an overflow in the respective vector component, and LB, MB and RB indicate the status of the left, middle and right mouse buttons (1 = pressed). PS/2 mice also understand several commands for reset and self-test, switching between different operating modes, and changing the resolution of the reported motion vectors.

	D7	D6	D5	D4	D3	D2	D1	D0
Byte1	YV	XV	YS	XS	1	MB	RB	LB
Byte2	X Movement							
Byte3	Y Movement							

Alternativ Pointing Device

- Trackball
- Touchpad
- Graphics Tablet
- Eye Ball Controlled



Thank You

